

2 - Root Locus method - Drawing rules

Root Locus drawing rules

$$\frac{D(s)G(s)}{1+D(s)G(s)} = \frac{K L(s)}{1+K L(s)}$$

Characteristic equation

roots of $1+K L(s)=0 \rightarrow$ closed loop poles

Varying K changes dynamic response

$$\left. \begin{aligned} 1+K L(s) &= 0 \\ 1+K \frac{b(s)}{a(s)} &= 0 \\ a(s)+K b(s) &= 0 \\ L(s) &= -1/K \end{aligned} \right\} \text{equivalent}$$

$$b(s) = \prod_{i=1}^m (s-z_i)$$

$$a(s) = \prod_{i=1}^n (s-p_i)$$

Key definitions

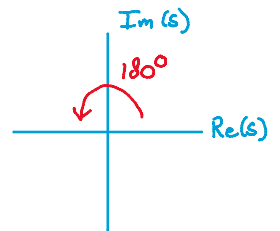
Definition I: $1+K L(s)=0$ must be satisfied under varying $K, K \in [0, \infty)$

$$L(s) = -1/K$$

$$Re^{st} = Re^{j\omega t}$$

$s=j\omega$ ← phase
← amplitude

$\mathbb{R}^+$



Definition II: Root locus of $L(s)$ is the set of points in the s-plane where the phase of $L(s)$ is 180°

l : integer value

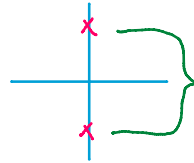
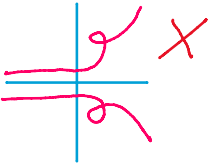
$$a(s) = s^n + a_1 s^{n-1} + \dots + a_n = 0$$

hard to solve...

easy to check!

$$\sum \psi_i - \sum \phi_i = 180^\circ + 360^\circ(l-1)$$

LTI system:



Complex conjugate behaviour

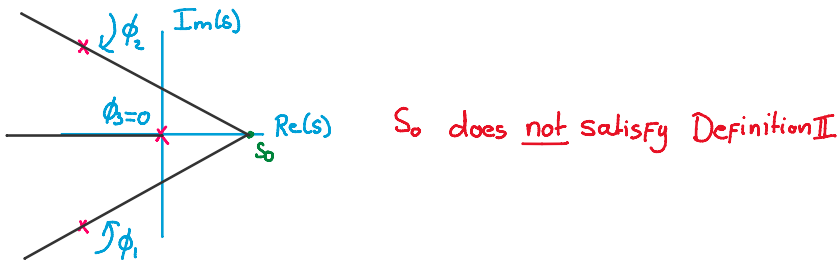
Drawing rules

Rule 1: n branches start at the poles of $L(s)$, m branches end at the zeros of $L(s)$

$$a(s) + K b(s) = 0 \Rightarrow \begin{cases} a(s) = 0 & K = 0 & \text{poles } (n) \\ b(s) = 0 & K \rightarrow \infty & \text{zeros } (m) \end{cases}$$

Strictly proper TF: $n-m$ branches go to ∞

Rule 2: The loci are on the real axis to the left of an odd number of poles and zeros



Rule 3: For large s and K , the loci that go to infinity are asymptotic lines radiating at a fixed angle from a central point, the centre of gravity of the poles.

$$\phi_l = \frac{180^\circ + 360^\circ(l-1)}{n-m} \quad l = 1, 2, \dots, n-m$$

loci going to infinity: $n-m$

angles at which the asymptotes branch out

$$\alpha = \frac{\sum p_i - \sum z_i}{n-m} \quad \text{only real parts needed!}$$

Complex conjugate pair: $\sum c \pm di = c + di + c - di = 2c$

Centre of gravity of the asymptotes

Intermediate example

$$G(s) = \frac{s+1}{(s-1)(s^2+4s-8)} \quad m=1 \quad z_i = -1$$

$$n=3 \quad p_i = 1, -2 \pm 2j$$

$$D(s) = 1$$

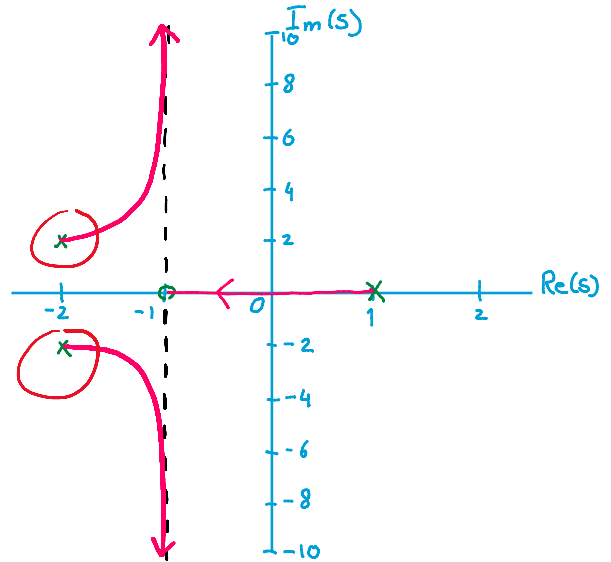
$$1 + D(s)G(s) = 1 + KL(s) = 0$$

$$\phi_l = \frac{180^\circ + 360^\circ(l-1)}{n-m} \quad l=1, n-m=2$$

$$n-m=2$$

$$\phi_1 = \frac{180^\circ + 0^\circ}{2} = 90^\circ \quad \phi_2 = \frac{180^\circ + 360^\circ}{2} = 270^\circ$$

$$\alpha = \frac{\sum p_i - \sum z_i}{n-m} = \frac{1-2-2-(-1)}{3-1} = -1$$



Rule 4: The departure angle of a branch of the locus from a single pole is given by

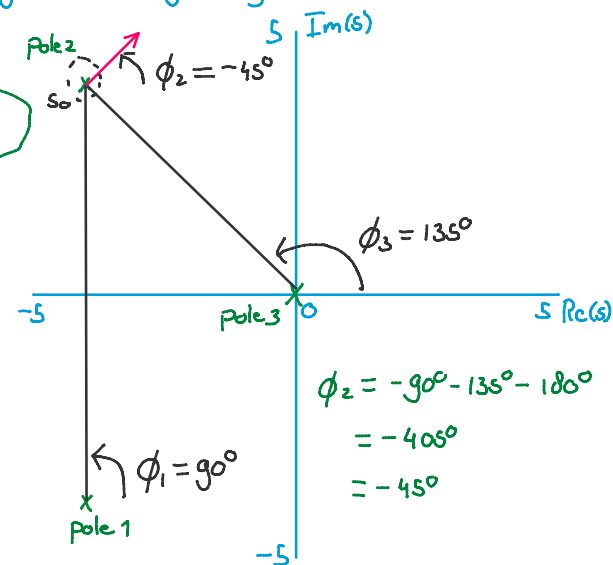
$$\phi_{\text{dep}} = \sum \psi_i - \sum_{i \neq \text{dep}} \phi_i - 180^\circ \quad \leftarrow \text{Definition II}$$

Sum of zeros
Sum of poles not considered

$$q \phi_{l, \text{dep}} = \sum \psi_i - \sum_{i \neq l, \text{dep}} \phi_i - 180^\circ - 360^\circ(l-1) \quad l=1, 2, \dots, q$$

q : repeated poles

$$q \psi_{l, \text{arr}} = \sum \phi_i - \sum_{i \neq l, \text{arr}} \psi_i + 180^\circ + 360^\circ(l-1) \quad l=1, 2, \dots, q$$



Rule 5: The locus can have multiple roots at points on the locus and the branches approach a point of q roots with angles separated by $\frac{180^\circ + 360^\circ(l-1)}{q}$

angles of departure and arrival: Continuation Locus

$$G(s) = \frac{s+2c}{s(s+c)} \quad 1 + KL(s) = 1 + K \frac{s+2}{s(s+1)} = 0$$

$$D(s) = 1$$

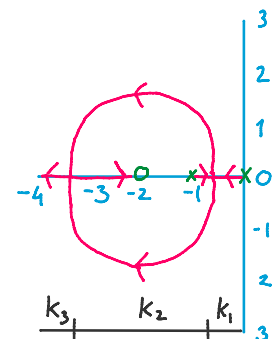
$$r_1, r_2 = \frac{-k+1 \pm \sqrt{k^2 - 6k + 1}}{2} \quad \text{Complex when } 0.172 < k < 5.828$$

$$K = K_1$$

$$K = K_1 + K_2$$

$$K = K_1 + K_2 + K_3$$

apply Rule 4!



Summary

* Drawing rules For Root Locus $\rightarrow$ understanding closed-loop behaviour through its open-loop plant

* Two definitions 1: $1 + KL(s) = 0$ $K \in \mathbb{R}^+$ 2: $\sum \psi_i + \sum \phi_i = 180^\circ + 360^\circ(\ell-1)$

* Rule 1: Branches from pole to zero or pole to infinity

Rule 2: Phase requirement, Loci on the real axis to the left of odd numbers of poles and zeros

Rule 3: Asymptotic behaviour (angles + central point) $\phi_\ell = \frac{180^\circ + 360^\circ(\ell-1)}{n-m}$ $\ell = 1, 2, \dots, n-m$

$$\sigma = \frac{\sum p_i - \sum z_i}{n-m}$$

Rule 4: Angles of departure and arrival

$$\angle \phi_{\ell, \text{dep}} = \sum \psi_i + \sum_{i \neq \ell, \text{dep}} \phi_i - 180^\circ - 360^\circ(\ell-1) \quad \angle \psi_{\ell, \text{arr}} = \sum_{i \neq \ell, \text{arr}} \psi_i + \sum \phi_i + 180^\circ + 360^\circ(\ell-1) \quad \ell = 1, 2, \dots, q$$

Rule 5: Multiple roots at point on the Locus $\frac{180^\circ + 360^\circ(\ell-1)}{q}$

$\hookrightarrow$ Continuation locus